

Intermittent Forecasting

24th June 2023



Agenda

- Introduction and explanation of models
 - Objective
 - Definition
 - Illustration
 - Number examples
- Case Studies
 - [Projects shortlisted](#)
 - [Visionworks](#), [AT&T Mexico](#) and Breuninger
 - EDA
 - Backtest cycles
 - Accuracy analysis
 - Examples
 - Inference
- [Takeaways and conclusion](#)
- References



Objective

- To evaluate models which are specifically designed for data with intermittent characteristics and summarize the observations.

Why did we start this initiative?

- Poor accuracy concerns from projects **VW for retail stores, GMG, Adidas, Breuninger** because of **intermittency** in their data.
- Though this segment is really difficult for any good accuracies we kept our objective to try to reduce the error by whatever percentage possible.

A Brief Explanation of The Models

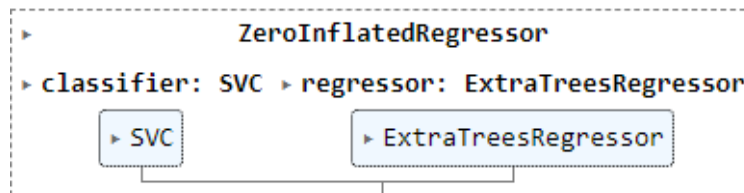
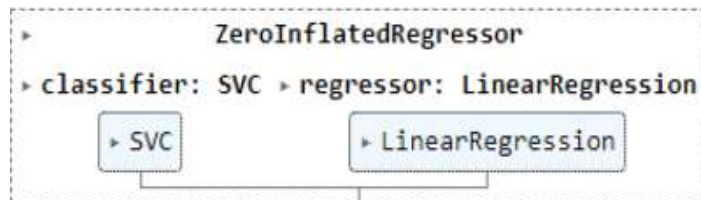


Zero Inflated Regression

Zero-inflated poisson regression is used to model **count data that has an excess of zero counts**

The zero-inflated regression model combines two components:

- **Inflation component:** Models the excess zeros. It typically uses a binary Classification algorithm to estimate the probability of observing a zero value versus a non-zero value. The predictors in this component help determine the presence or absence of excess zeros.
- **Regression component:** Models the non-zero values. It uses a count distribution, such as the Poisson or negative binomial distribution or any regression models, to model the count data when the response is non-zero. The predictors in this component help explain the relationship between the predictors and the non-zero values.



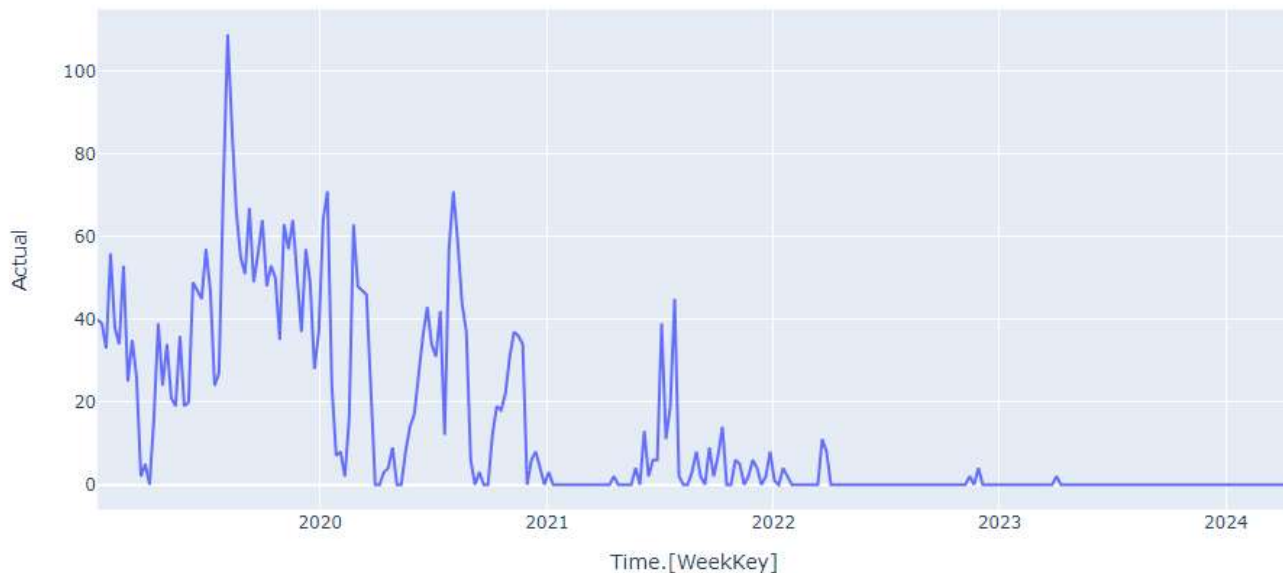
****Please [click here](#) for more information.**

Let's understand with an example



Let's take an intermittent time series here: The intersection
1921-560

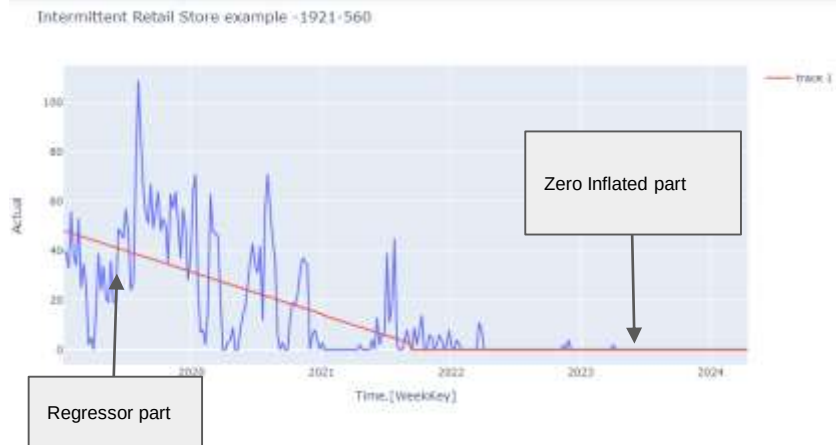
Intermittent Retail Store example -1921-560



After fitting the model

Taking predictor variable as weekno (1,2,3,.. 52)

```
#Model fit
model= zir.fit(Zerofilled_example_2[['Weekno']],Zerofilled_example_2.Actual)
```

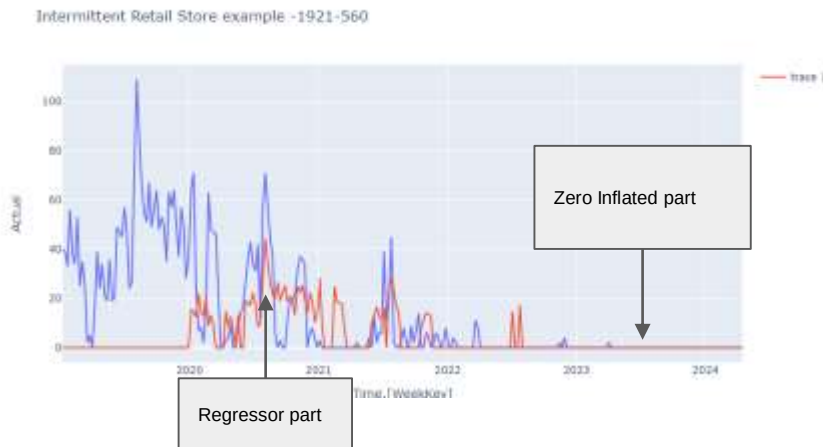


```
In [158]: classification_Accuracy_roc_auc_score
```

```
Out[158]: 0.6301068510370836
```

Using predictor variables as actuals lag (Eg., Lag 3, Lag 12)

```
#model-fit
model2 = zir.fit(Zerofilled_example_2[['Lag_variable']],Zerofilled_example_2.Actual)
```





What is Periodicity?

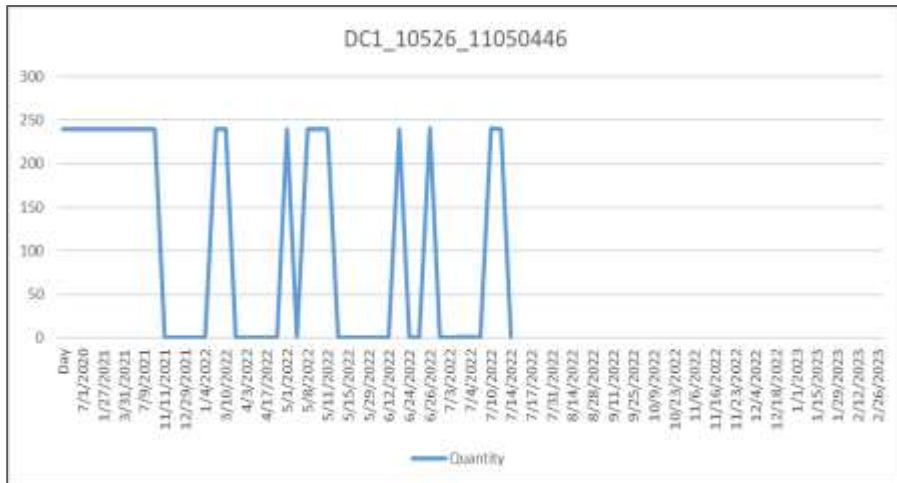
- Periodicity (Average Demand Interval) is the time interval between two positive demand points.
- Periodicity can be measured either by using the transaction with the positive points or the quantities that are shipped.

What is Periodicity Model?

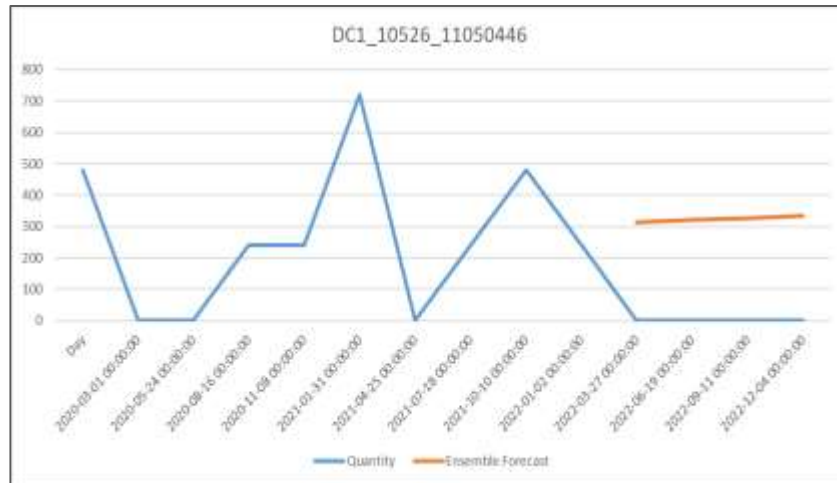
- Calculates the Periodicity for the given time series
- Aggregates our time Series data to the Periodicity level
- Forecasts at the aggregated level
- Disaggregates the forecast to the desired lower time bucket level from periodicity level.

Number Example | Periodicity Model

Example from Starbucks : **DC1_10526_11050446**



Data After Aggregation at Periodicity level (Forecast level)



combined	avg_periodicity_dates	median_periodicity_dates	std_periodicity_dates	mode_periodicity_dates
DC1_10526_11050446	12	8	12	12

- Forecast generation time bucket : Weeks.
- We can then disaggregate the forecasts to lower level by dividing by 12.(the mean periodicity).

- Aggregate-Disaggregate Intermittent Demand Approach: Uses temporal aggregation to reduce the number of zero observations. Once the data has been aggregated, it uses the optimized SES to generate the forecasts at the new level. It then breaks down the forecast to the original level using equal weights.
- This model is from the Nixtla stats-forecast library.

Month	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12			
Original series	1	0	0	0	5	6	0	3	2						
Aggregated series	1			11			5			7.4			Prediction using SES		
Final Prediction										2.47	2.47	2.47	Dissag by equal weights		

- Intermittent Multiple Aggregation Prediction Algorithm: Similar to ADIDA, but instead of using a single aggregation level, it considers multiple in order to capture different dynamics of the data. Uses the optimized SES to generate the forecasts at the new levels and then combines them using a simple average.
- This model is from the Nixtla stats-forecast library.

Month	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14	M15				
Original Series	1	0	0	0	5	6	0	3	2	0	0	3							
Aggregated Series 1	1			11			5			3			3.8					SES Prediction	
Aggregated Series 2	1				14				5				8.6						Simple avg of series 1 and 2 then disagg
Final Prediction													1.71	1.71	1.71	1.71			



- This model is from the Nixtla statsforecast library.

Teunter-Syntetos-Babai: A modification of Croston's method that replaces the inter-demand intervals with the demand probability d_t , which is defined as follows:

$$d_t = \begin{cases} 1 & \text{if demand occurs at time } t \\ 0 & \text{otherwise.} \end{cases}$$

Hence, the forecast is given by

$$\hat{y}_t = \hat{d}_t \hat{z}_t$$

Both d_t and z_t are forecasted using SES. The smoothing parameters of each may differ, like in the optimized Croston's method.

Projects shortlisted for case study



VW

- Industry Verticals - RDAF
- Sub-Industry - RD
- Significant lag - Weekly Lag 4
- **Forecast level** - Item Class x Location x Week
- **Horizon** - 52 weeks
- **Backtest Cycles** - 4
- Backtest Cycle
 - Current week dates - W03-2023,W04-2023, W05-2023,W06-2023.

AT&T

- Industry Verticals - RDAF
- Sub-Industry - Telco Retail
- Significant lags - Monthly lag 1
- **Forecast level** - Model memory X Sales Channel X Planning Month
- **Horizon** - 6 months
- **Backtest cycles** - 3
- Backtest current months - Nov-22, Dec-22, Jan-23

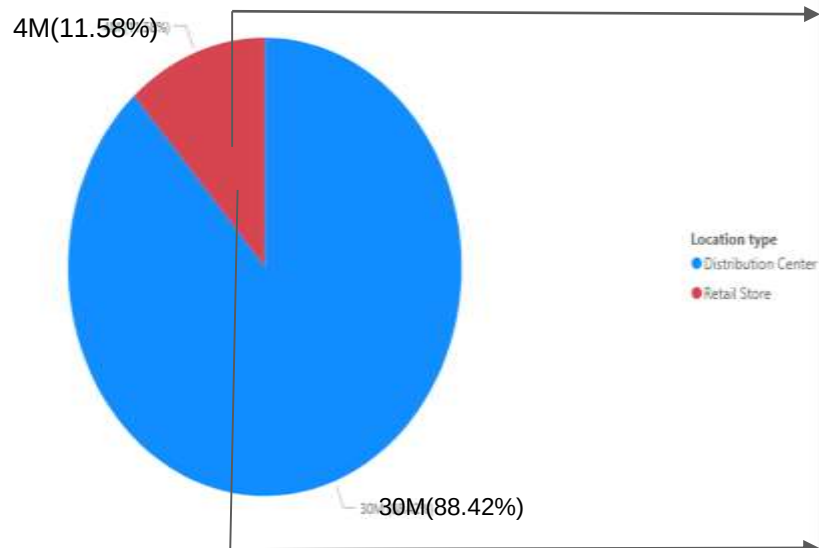
Breuninger

- Industry Verticals - RDAF
- Sub-Industry - AFL
- Significant lags - Monthly Lag1
- **Forecast level** - Planning Item x Locationx Month Level
- **Forecast Horizon** - 10 months
- **Backtest cycles** - 2
- Backtest Current Months - April 2023, May-2023.

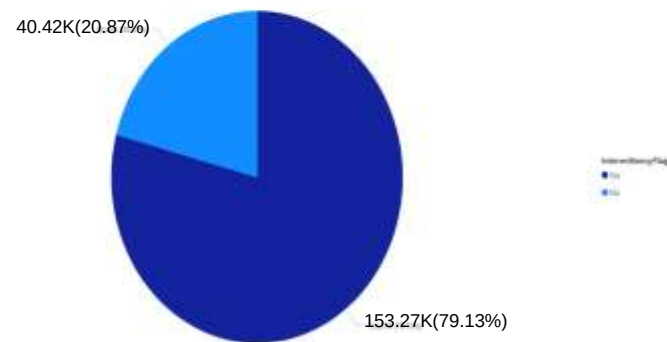
Case Studies : VisionWorks



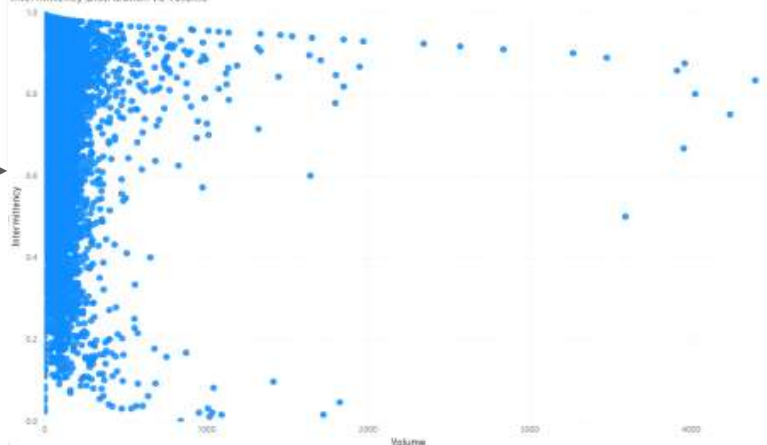
Case Study | VisionWorks | Brief EDA



Intersection intermittency flag



Interimittency Distribution vs Volume



The intersections present in retail stores have very low **rate of sales** ≤ 0.2 and are also **highly intermittent**. Hence we are taking retail stores into consideration for case studies in VisionWorks.

Case Study | VisionWorks | Backtest Cycle Plan



W03-2023	W04-2023	W05-2023	W06-2023	W07-2023	W08-2023	W09-2023	W10-2023
W03-2023	W04-2023	W05-2023	W06-2023	W07-2023	W08-2023	W09-2023	W10-2023
W03-2023	W04-2023	W05-2023	W06-2023	W07-2023	W08-2023	W09-2023	W10-2023
W03-2023	W04-2023	W05-2023	W06-2023	W07-2023	W08-2023	W09-2023	W10-2023

Current Week

Lag 4

Accuracy Analysis | Retail Stores

Demand Class	Count of Intersections	%Volume-Contribution	Average of CoV	QTY	Periodicity-Model	Zir_sklego (lag 8)	Zir_sklego (lag 13)	Zir_sklego (lag 52)	ZIR_Sm	o9_Stat	ADIDA	imapa	tsb
AX	6	0.97%	0.44	4385	70.52%	73.34%	73.34%	65.25%	29.83%	66.56%	71.39%	71.39%	67.88%
AY	23023	91.49%	2.29	412686	-9.65%	-7.93%	-7.83%	-13.83%	-23.57%	-23.98%	-4.55%	-4.58%	-5.65%
BY	11211	5.99%	5.12	27035	-41.57%	-37.26%	-37.02%	-39.06%	-35.37%	-42.34%	-35.94%	-36.03%	-37.89%
NEW LAUNCH	2332	1.54%	2.76	6945	-31.81%	-25.39%	-25.28%	-32.21%	-31.56%	-14.29%	-24.50%	-24.54%	-25.40%
Grand Total	36572	100.00%	3.19	451051	-13.83%	-11.66%	-11.55%	-17.00%	-24.9%	-23.53%	-8.67%	-8.70%	-9.90%

Observations :

- ZIR sklego and Periodicity models perform better than the DPRef Forecasting algorithms already present.
- ADIDA, iMAPA and TSB perform better than ZIR sklego and Periodicity Model.



Demand Class	Count of Intersection	Average of CoV	QTY	Periodicity	Zir_sklego (lag 13)	Zir_sklego (lag_8)	Zir_sklego (lag 52)	ZIR_sm	o9_Stat	ADIDA	imapa	tsb
AY	21143	2.38	284876	-26.5%	-23.4%	-23.3%	-28.68%	-31.2%	-37.4%	-20.1%	-20.12%	-21.3%
BY	11208	5.12	27029	-42.0%	-37.4%	-37.1%	-39.43%	-35.7%	-42.3%	-36.3%	-35.39%	-38.3%
NEW LAUNCH	1266	2.93	5183	-46.5%	-37.4%	-37.0%	-45.40%	-49.0%	-16.7%	-36.4%	-34.41%	-37.5%
Grand Total	33617	3.32	317088	-29.8%	-26.4%	-26.2%	-30.98%	-32.2%	-37.3%	-23.5%	-23.52%	-24.8%

Observations :

- ZIR sklego and Periodicity models perform better than the DPRef Forecasting algorithms already present.
- ADIDA, iMAPA and TSB perform better than ZIR sklego and Periodicity Model.

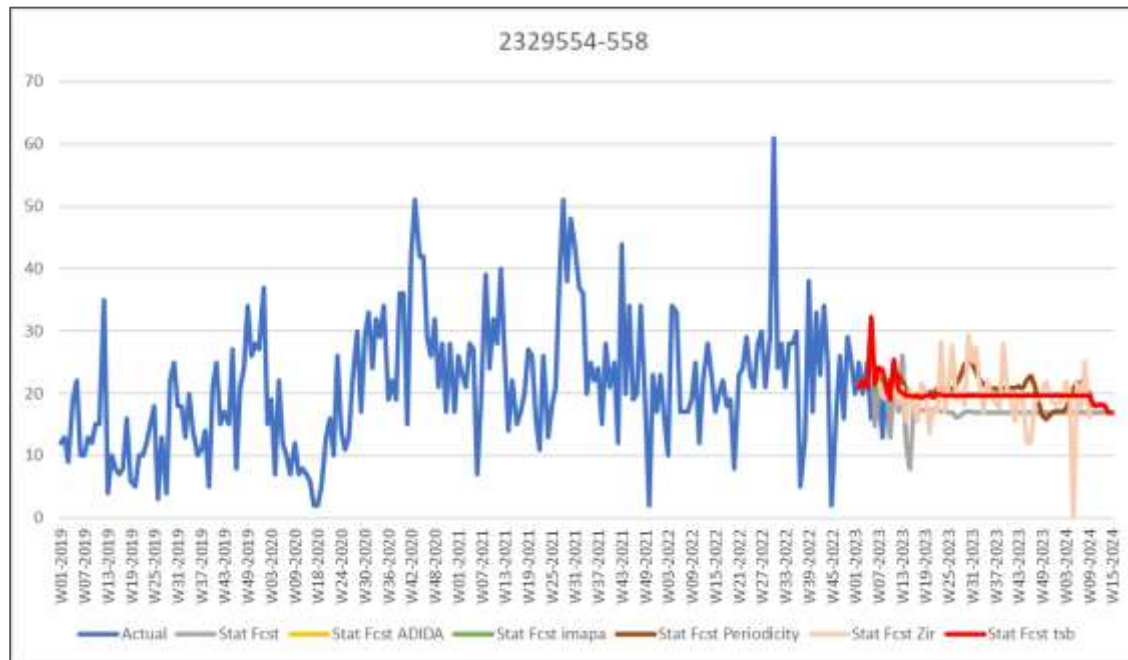


Demand Class	Count of Intersection	Average of CoV	QTY	Periodicity_ MAPE	ZIR_sklego (Lag 8)	Zir_sklego (Lag52)	imapa	tsb	ADIDA	o9_Stat
AY	21143	2.38	284876	126.5%	123.4%	128.7%	120.1%	121.3%	120.1%	137.4%
BY	11208	5.12	27029	142.0%	137.4%	139.4%	136.4%	138.3%	136.3%	142.3%
NEW LAUNCH	1266	2.93	5183	146.5%	137.4%	145.4%	136.4%	137.5%	136.4%	116.7%
Grand Total	33617	3.32	317088	129.8%	126.4%	131.0%	123.5%	124.8%	123.5%	137.3%

Observations :

- ZIR sklego and Periodicity models perform better than the DPRef Forecasting algorithms already present.
- ADIDA, iMAPA and TSB perform better than ZIR sklego and Periodicity Model.

Example | 2329554-558 (Lens)



Future Forecast period : 52

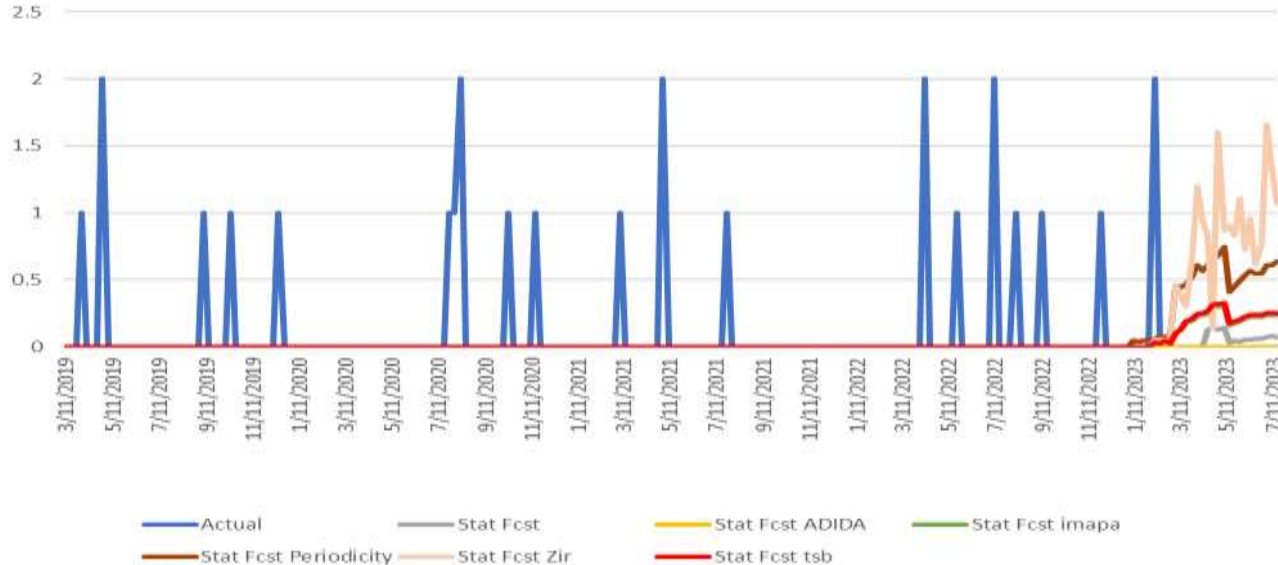
Future Forecast start week:
W11-2023

Observations: Here Zir sklego Forward Forecasts show a level match forecast to the actuals compared to other forecasts The ADIDA tsb and imapa lines almost overlap give flat lines and overlap with each other.

Examples-VW-1148029-583(Frames)



1148029_All_583



Future Forecast
period:52

Future Forecast
start week:W11-
2023

Observations:The sklego-zir and periodicity model try to capture the mean pattern from the recent history whereas the ADIDA forecast gives an almost 0 (which is rounded to 0) line for prediction horizon.

Case Study | VisionWorks USA : Retail Stores | Iteration Summary



Model Name	Model Type	Features Used	Value Added	Comments
ADIDA	Statistical	None	👉	Accuracy better than DPref - Better MAPE
Croston(present in DPref)	Statistical	None	➖	Already present in DPref
IMAPA	Statistical	None	👉	Accuracy better than DPref - Least MAPE
TSB	Statistical	Only hyperparameters	👉	Accuracy better than DPref
ZIR-sklego	ML (SVC and ExtraTreesRegressor)	1st iteration-lag 8 2nd iteration -lag 13 3rd iteration-lag 52	👉	Accuracy better than DPref
Periodicity Model	Ensemble(prophet,SES,moving average)	None	👉	Accuracy better than DPref
ZIR Statsmodel(Inflated poisson Regression)	Ensemble(logistic regression and Poisson Regression)	1st iteration-lag 8 2nd iteration -lag 13 3rd iteration-lag 52	➖	Accuracy worse than DPref models

Script Run time for full scope is ~45 mins + disaggregation procedure ~45 mins

Minimum data points for each model is 2

Case Studies : AT&T Mexico



Case Study - AT&T Mexico | Segmentation/Brief EDA



Volume-Variability	Count of tsid
AX	112
AY	524
BX	1352
BY	2546
Grand Total	4534

Intermittency	Count of tsid	Percent age
NO	2023	44.6%
YES	2511	55.4%
Grand Total	4534	100%

PLC Status	Count of tsid
DISC	1438
MATURE	1901
NEW LAUNCH	1195
Grand Total	4534

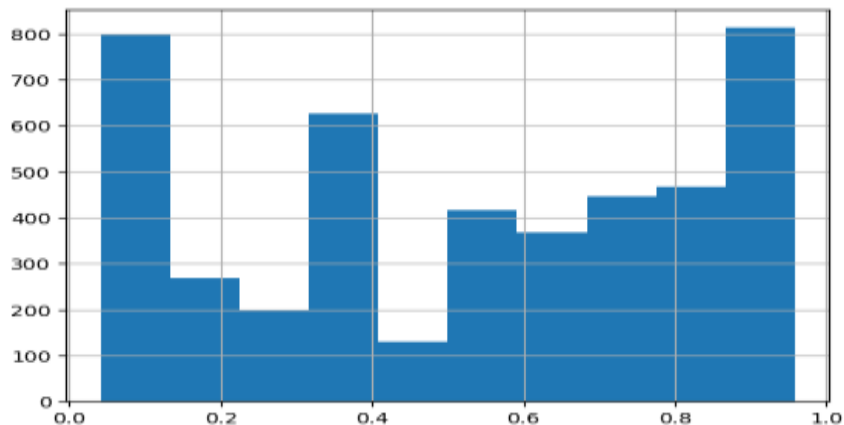


Fig - Distribution of zero ratios which is the number of zero sales to the length of time-series.

Backtest Cycles For AT&T Mexico



M10-2022	M11-2022	M12-2022	M1-2023	M2-2023	M3-2023	M4-2023
M10-2022	M11-2022	M12-2022	M1-2023	M2-2023	M3-2023	M4-2023
M10-2022	M11-2022	M12-2022	M1-2023	M2-2023	M3-2023	M4-2023

Current Week

Lag 4

Accuracy Analysis | AT&T Mexico

Year	Month	Accuracy adida fcst	Accuracy crostonoptim fcst	Accuracy imapa fcst	Accuracy tsb fcst	Accuracy ZIR fcst	Accuracy ZIR gbm withoutlags fcst	Accuracy ZIR RF withoutlags fcst	Accuracy Stat Fcst L1
22	11	23%	31%	30%	53%	27%	34%	34%	32%
22	12	39%	44%	44%	67%	36%	58%	36%	40%
23	1	21%	28%	28%	50%	22%	27%	26%	23%
23	2	30%	36%	38%	63%	29%	25%	25%	22%
Grand Total		10%	-14%	15%	28%	7%	12%	12%	2%

Year	Month	Accuracy adida fcst	Accuracy crostonoptim fcst	Accuracy imapa fcst	Accuracy tsb fcst	Accuracy ZIR fcst	Accuracy ZIR gbm withoutlags fcst	Accuracy ZIR RF withoutlags fcst	Accuracy Stat Fcst L1
22	12	22%	27%	36%	63%	80%	47%	47%	5%
23	1	27%	35%	42%	65%	70%	42%	42%	-3%
23	2	27%	33%	43%	67%	79%	40%	39%	-5%
23	3	4%	-48%	3%	-6%	-13%	-5%	-5%	-14%
Grand Total		10%	-31%	15%	19%	20%	13%	13%	-14%

Year	Month	Accuracy adida fcst	Accuracy crostonoptim fcst	Accuracy imapa fcst	Accuracy tsb fcst	Accuracy ZIR fcst	Accuracy ZIR gbm withoutl ags fcst	Accuracy ZIR RF withoutl ags fcst	Accuracy Stat Fcst L1
23	1	-495%	-7147%	-451%	-124%	-4299%	-4557%	-4296%	-6841%
23	2	-554%	-8003%	-505%	-140%	-4822%	-5104%	-4812%	-7194%
23	3	-88%	-1195%	-84%	-37%	-750%	-475%	-569%	-931%
23	4	-130%	-1643%	-124%	-60%	-861%	-128%	-136%	-562%
Grand Total		-173%	-2371%	-161%	-60%	-1396%	-1044%	-1049%	-1770%

Observation: ZIR, TSB and iMAPA are best models for intermittent time series in terms of accuracy.

MAE Analysis | AT&T Mexico

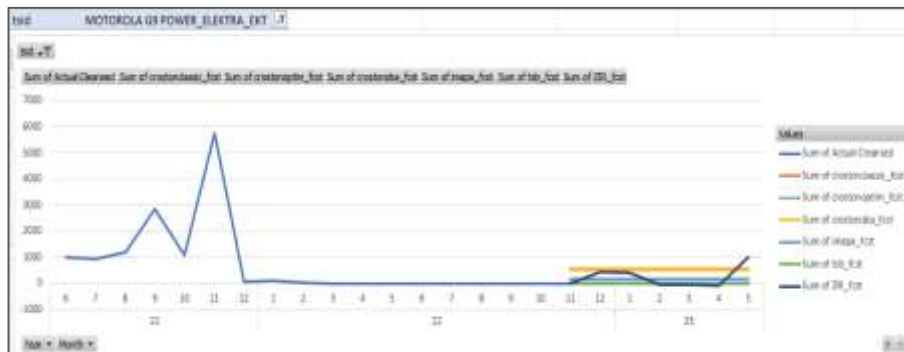
Year	Month	MAE adida fcst	MAE crostonoptim fcst	MAE imapa fcst	MAE tsb fcst	MAE ZIR fcst	MAE ZIR gbm withoutlags fcst	MAE ZIR RF withoutlag s fcst	MAE o9 stat fcst
22	11	55	49	50	33	52	47	47	49
22	12	53	49	48	29	56	37	37	53
23	1	75	69	69	48	75	70	70	73
23	2	59	53	52	31	60	63	63	65
23	3	83	117	82	84	89	90	90	93
23	4	59	107	59	60	64	59	59	77
Grand Total		67	85	63	55	69	65	65	73

Year	Month	MAE adida fcst	MAE crostonoptim fcst	MAE imapa fcst	MAE tsb fcst	MAE ZIR fcst	MAE ZIR gbm withoutlags fcst	MAE ZIR RF withoutlags fcst	MAE o9 stat fcst
22	12	29	27	24	14	8	19	19	35
23	1	33	29	26	16	13	25	25	45
23	2	29	27	23	14	9	24	24	42
23	3	60	92	60	66	70	65	65	70
23	4	54	98	54	55	55	54	55	72
Grand Total		46	66	43	41	40	44	44	58

Year	Month	MAE adida fcst	MAE crostonoptim fcst	MAE imapa fcst	MAE tsb fcst	MAE ZIR fcst	MAE ZIR gbm withoutlags fcst	MAE ZIR RF withoutlags fcst	MAE o9 stat fcst
23	1	5	64	5	2	39	41	39	61
23	2	5	64	5	2	39	41	39	57
23	3	10	65	9	7	43	29	34	52
23	4	9	65	8	6	36	9	9	25
Grand Total		7	65	7	4	39	30	30	49

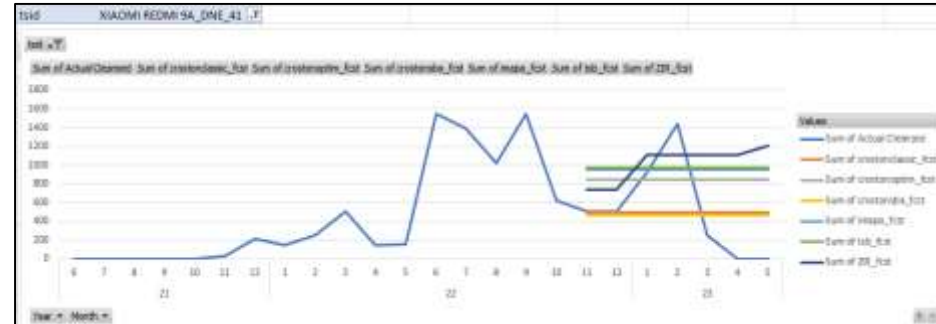
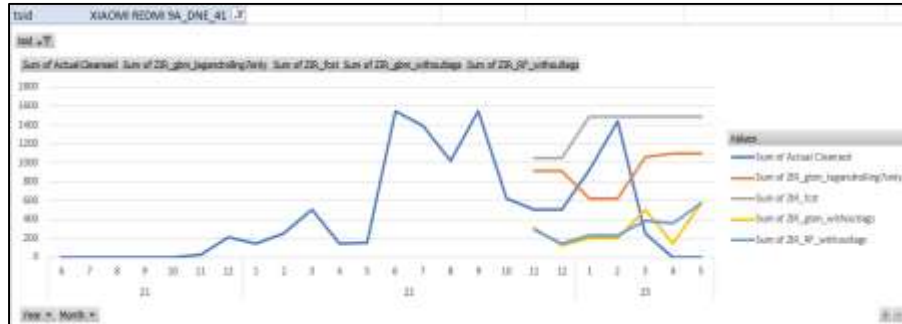
Observation: ZIR, TSB and iMAPA are best models for intermittent time series in terms of MAE also.

Example



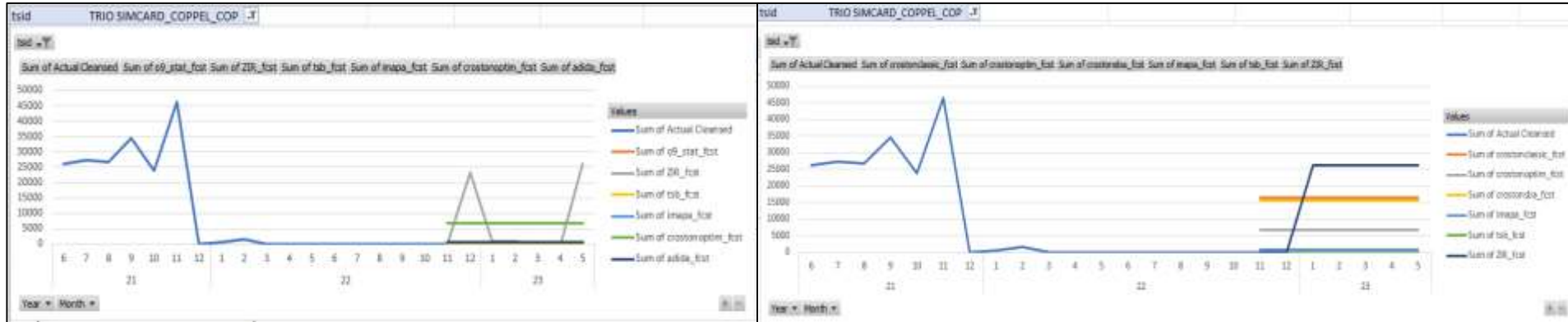
We observe that in the left figure for the extreme intermittent timeseries, the statistical models perform better as compared to ZIR whereas in the right figure where the number of zeros are less, the ZIR regressor helps to capture the data pattern from history better than the other models

Example for tsid XIAOMI REDMI 9A_DNE_41 Zero Inflated Model analysis



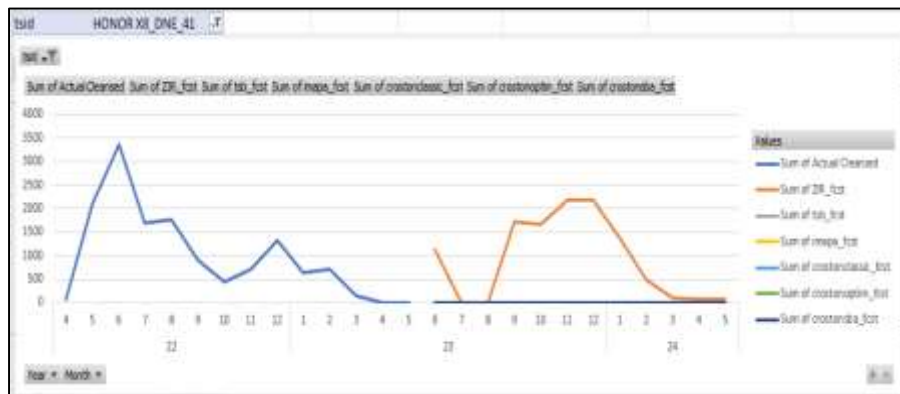
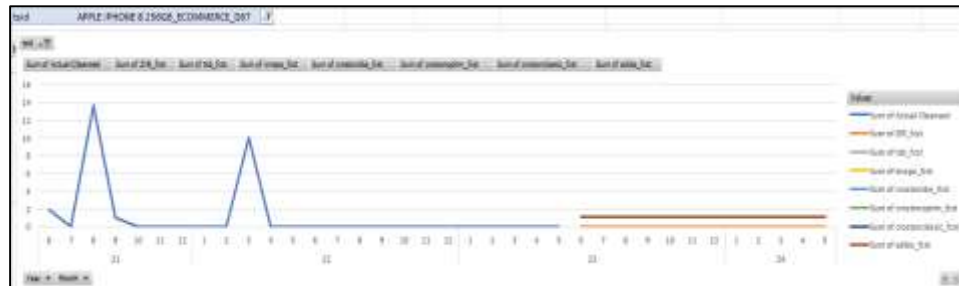
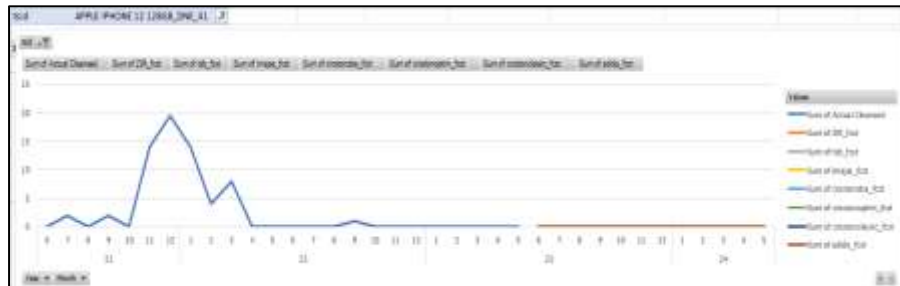
ZIR works well in capturing temporal and time embedded (lag) features from the history due to the ML regressor counterpart which gives it a upper hand for non-intermittent time-series also.

Example for tsid TRIO SIMCARD_COPPEL_COP

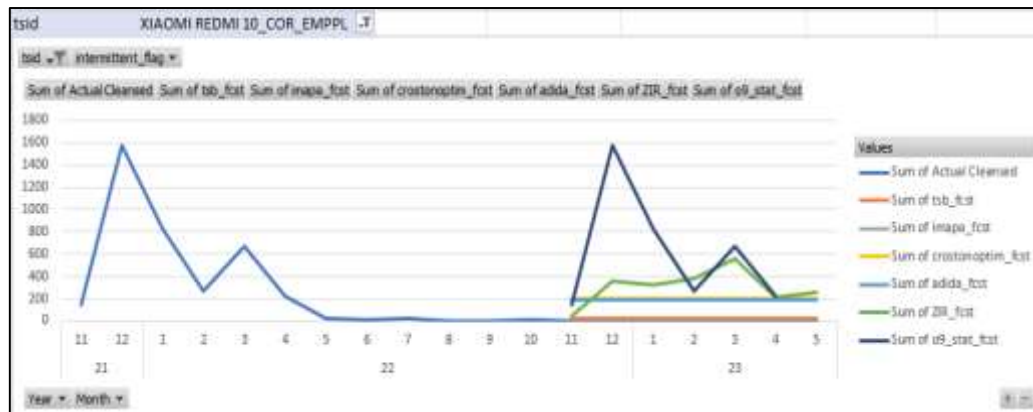
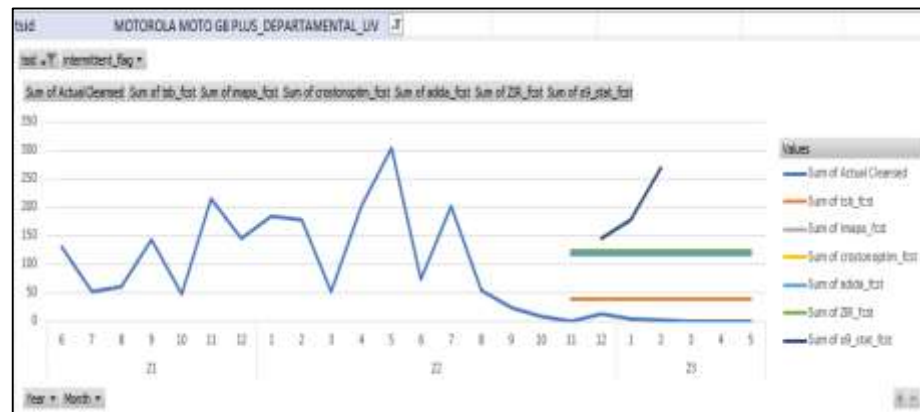


Above fig. shows ZIR with *lag, rolling, month and qtr number* (left) and *lag, rolling and month number* (right) along with other models. Although in some cases the ZIR tries to mimic the seasonal nature from the history, the other statistical models such as ADIDA, IMAPA, Croston TSB works well in capturing the levels of the data due to their advanced nature tweaking the SES algorithm. We can tweak the performance of ZIR by choosing the appropriate features for the fitted regressor model.

Example of forward forecasts



Example tsids with better forecasts than ref forecasts



Case Study | AT&T Mexico | Iteration Summary

Model Name	Model Type	Features Used	Value Added	Comments
ADIDA	Statistical	None	👉	Accuracy better than DPRef for 2 backtest cycles
Croston(present in DPRef)	Statistical	None	⊖	Already present in DPRef
IMAPA	Statistical	None	👉	Accuracy better than DPRef for 2 backtest cycles
TSB	Statistical	Only hyperparameters	👉	Accuracy better than DPRef for 2 backtest cycles
ZIR-sklego	ML (SVC and GradientBoostingRegressor or RandomForestRegressor)	1st iteration-lag and rolling 7, monthno. 2nd iteration - RF without lagandrolling 3rd iteration- GBM without lagandrolling	👉	Accuracy better than DPRef

Script Run time for full scope is ~10 mins

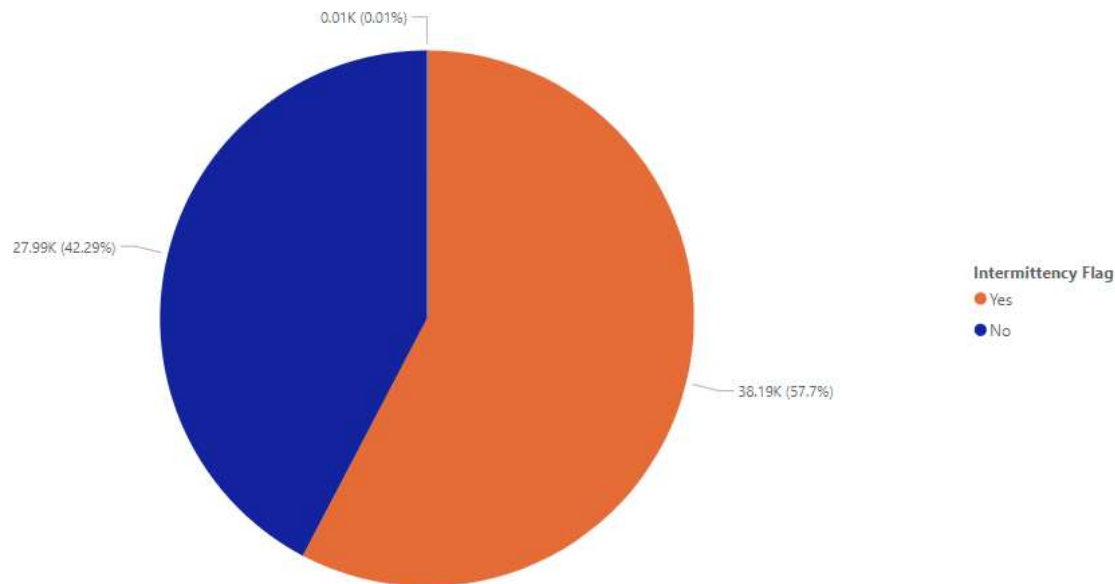
Minimum data points for each model is 2

*Case Studies : Breuninger
(WIP)... validation is
ongoing..*



Case Study | Breuninger | Brief Intermittency EDA

Count of ts_id by Intermittency Flag





M03-2023	M04-2023	M05-2023	M06-2023
M03-2023	M04-2023	M05-2023	M06-2023

Current Month

No of cycles : 2

Lag 1

Backtest cycle window: Month
04-2023 to Month 05-2023

Accuracy Analysis on Breuninger data from BT1 and BT2

Months	ADIDA	imapa	tsb	ZIR_sklego	crostonclassic (DPreRef)	Periodicity
Apr	29.1%	29.5%	37.3%	28.1%	11.8%	25.7%
May	27.5%	27.9%	39.0%	29.1%	12.2%	28.1%
Grand Total	28.2%	28.6%	38.3%	28.7%	12.1%	27.0%

Mention the scope of data

Intersections - ~8000 active time series

NOS -

NPIs are not part of this? - confirm?(confirmed)

There should be considering your intermittent segment of data from Breuninger

Observations: Accuracy of ADIDA, imapa ,tsb and ZIR are better than the croston model which is currently present in DPreRef.

Accuracy Analysis | Intermittency Flag :Yes



Intermittency_Flag	Yes					
Months	ADIDA	imapa	tsb	ZIR_sklego	crostonclassic (DPRef)	Periodicity
+ Apr	-62.7%	-57.0%	-18.9%	-31.6%	-205.6%	-67.1%
+ May	-60.5%	-51.1%	-13.5%	-31.8%	-198.8%	-65.6%
Grand Total	-61.6%	-54.0%	-16.2%	-31.7%	-202.1%	-66.3%

Observations:Accuracy of ADIDA, imapa ,tsb and ZIR are better than the croston model which is currently present in DPRef.

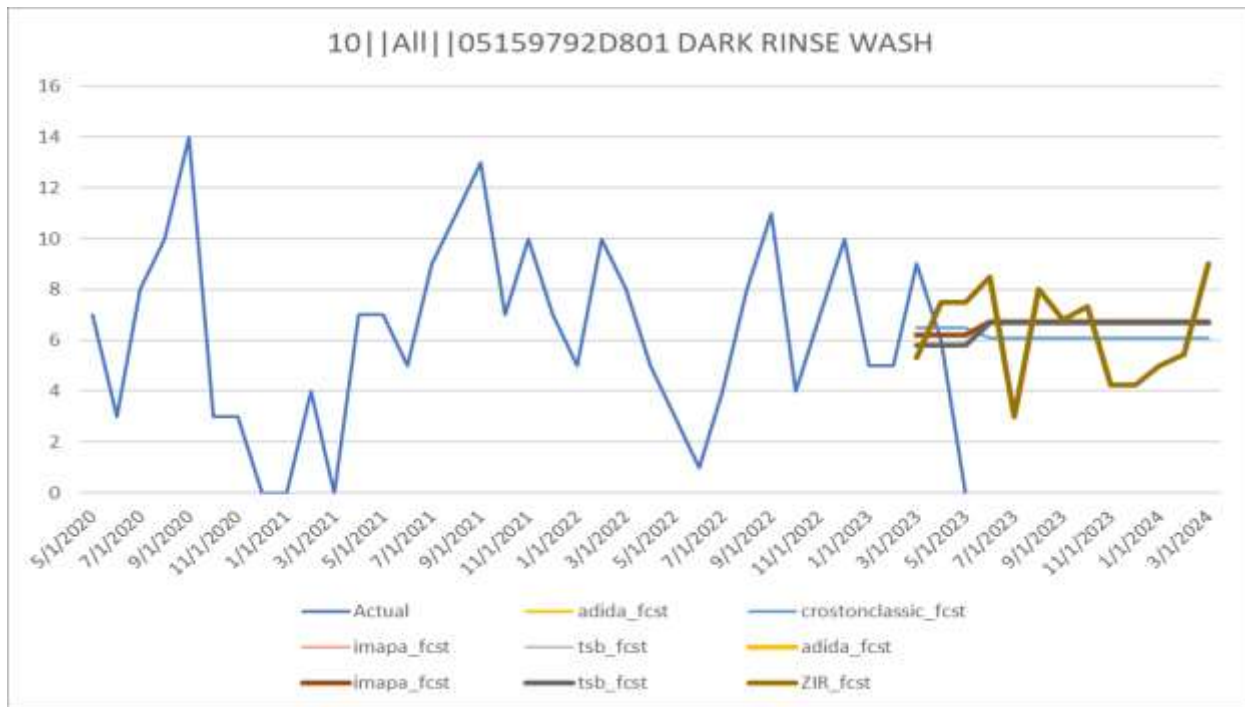
Case Study | Breuninger | Iteration summary



Model Name	Model Type	Features Used	Value Added	Comments
ADIDA	Statistical	None	➡	Accuracy better than DPreRef(Croston)
Croston(present in DPreRef)	Statistical	None	⊖	Present in DPreRef
IMAPA	Statistical	None	➡	Accuracy better than DPreRef(Croston)
TSB	Statistical	Only hyperparameters	➡	Accuracy better than DPreRef (Croston)
ZIR-sklego	ML (SVC and ExtraTreesRegressor)	lag 3	➡	Accuracy better than DPreRef
ZIR Statsmodel(Inflated poisson Regression)	Ensemble(logistic regression and Poisson Regression)	lag 3	⊖	Accuracy worse than DPreRef
Periodicity	Ensemble(prophet,SES, moving average)	None	➡	Accuracy better than DPreRef

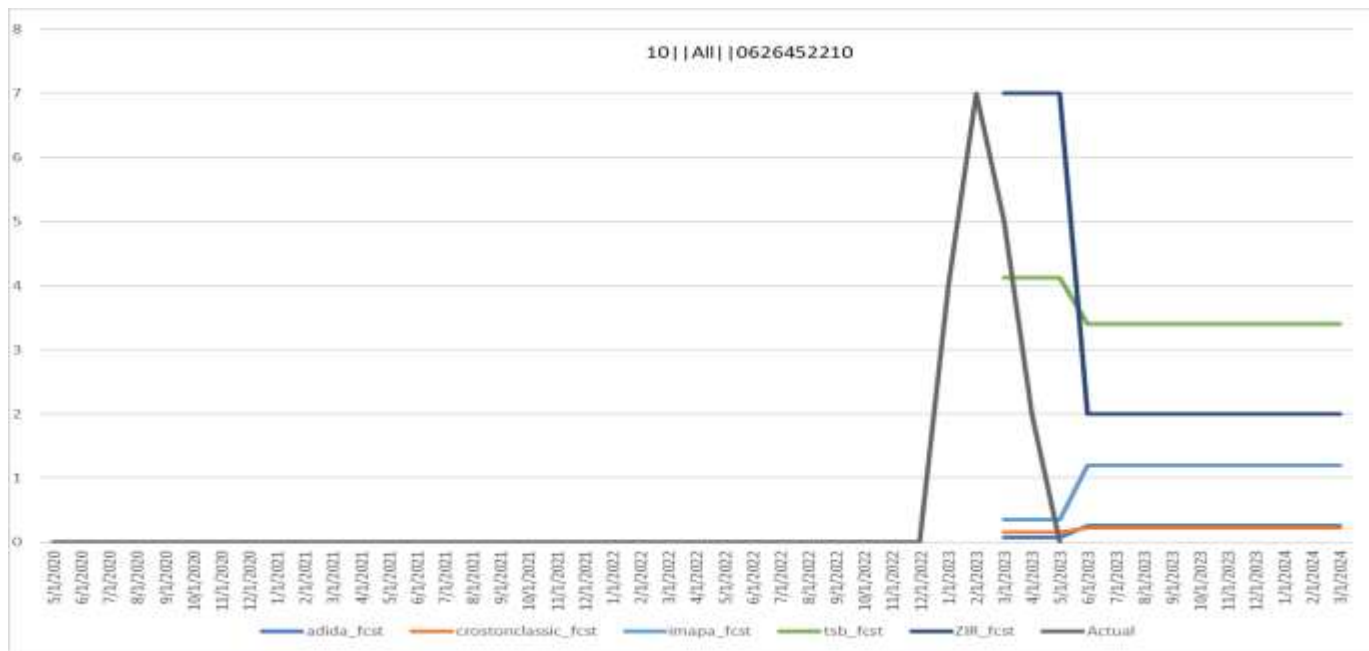
Runtime taken for full scope (active items)~40 mins.

Example 1 : 10||All||0515972D801 DARK RINSE WASH



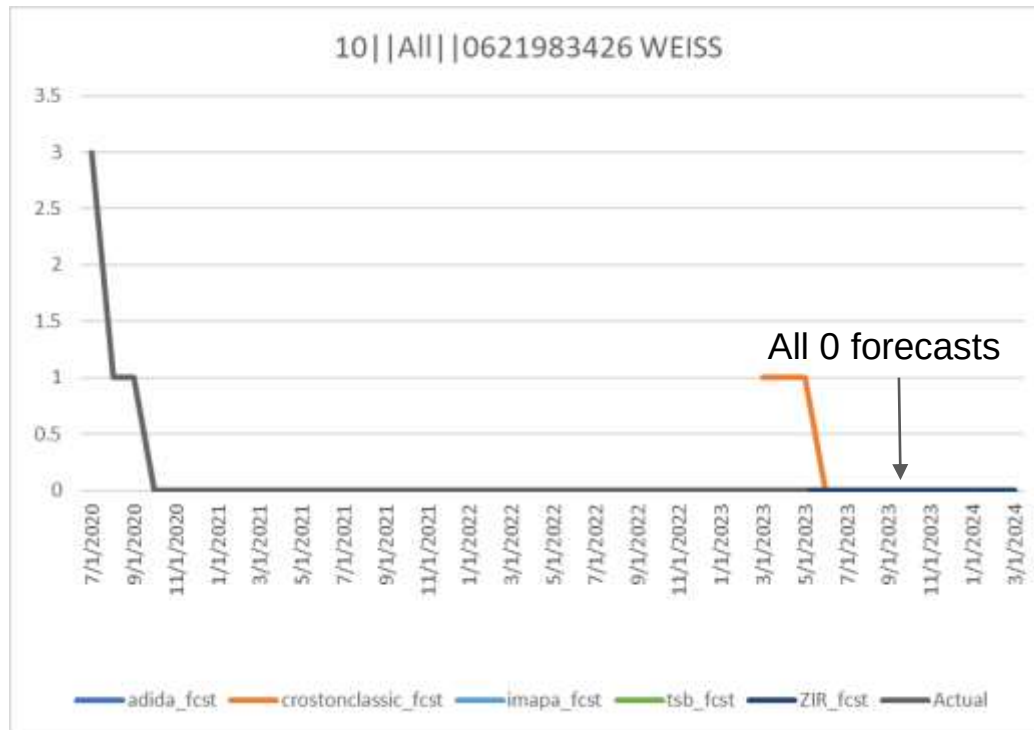
Observations: The ZIR_forecasts and the statmodel forecasts seem to be better than the normal croston forecasts/DPRef Forecasts.

Example 2 : 10||All||0626452210



Observations: The crostonclassic and adida forecasts give almost 0 forecasts when compared to other models which try to capture the forward forecast patterns in the range of 1-4 units. The models closer to 0 seem to be more accurate than the others models giving higher forecast.

Example 3 : 10||All||0621983426 WEISS



Observations: Almost all forecasts give 0 forecast except for croston classic which tries to capture based on the pattern in 2020.

Accuracy at week level | Total combined(3 months)



Time.[WeekKey]	adida_disagg	imapa_disagg	tsb_disagg	ZIR_disagg	crostonclassic
4/3/2023	30.8%	31.1%	36.3%	32.7%	27.6%
4/10/2023	31.6%	31.7%	36.6%	33.9%	29.5%
4/17/2023	32.7%	32.8%	37.0%	34.6%	30.0%
4/24/2023	34.7%	34.9%	38.9%	36.2%	32.9%
5/1/2023	24.7%	24.7%	29.8%	24.5%	22.1%
5/8/2023	27.6%	27.7%	30.6%	29.5%	25.9%
5/15/2023	26.3%	26.4%	31.1%	27.3%	23.3%
5/22/2023	26.6%	26.6%	30.2%	29.0%	26.0%
5/29/2023	25.6%	25.7%	29.8%	28.2%	23.9%
6/5/2023	-28.0%	-28.1%	-37.6%	-38.6%	-21.1%
6/12/2023	-25.0%	-24.9%	-31.2%	-35.6%	-21.6%
Grand Total	23.2%	23.3%	26.4%	23.5%	21.8%

Takeaways and Inferences



Client	Inference
Visionworks : Retail Stores	The statsmodels including <i>ADIDA</i> , <i>iMAPA</i> , <i>TSB</i> and models like ZIR and Periodicity are better performing compared to the models in DPRef. Lag features are the only features used in ZIR.
AT&T Mexico	<i>ADIDA</i> , <i>iMAPA</i> , <i>TSB</i> and <i>ZIR</i> are good models for intermittent time series in terms of accuracy and reasonably good runtimes. The <i>ZIR</i> model can be customized to give better results by using engineered features to capture data patterns from history.
Breuninger	For Breuninger, the statsmodels like <i>ADIDA</i> , <i>iMAPA</i> , <i>TSB</i> and models like ZIR are used and they give better accuracy than croston-classic used in DPref. Only lag features have been used for ZIR.

Assumptions



Future Scope



References



- <https://towardsdatascience.com/zero-inflated-regression-c7dfc656d8af>-Zero inflated Regression using sklego
- <https://timeseriesreasoning.com/contents/zero-inflated-poisson-regression-model/>- Zero inflated poisson Regression present in statsmodels.
- <https://www.hindawi.com/journals/mpe/2021/7001710/>- Paper on ExtraTreeRegressor.
- <https://nixtla.github.io/statsforecast/models.html> - Nixtla documentation
- TSB

Thank you!



Demand Planning

Appendix

Zero inflated model classification score (ROC area score)

ts_id	ZIR_roc_auc
17 All 0699574491 LIGHT INDIGO - WORN IN	0.944444444
50-DE All 0715584826682 DEEP SATEEN GLAM	0.9375
50-DE All 07054346BLACK	0.928571429
14 All 0709144902 BLACK	0.906832298
10 All 0707007490 LIGHT INDIGO - WORN IN	0.9
50-DE All 0705874350023 PINE GREEN	0.857142857
50-DE All 070997441030 DARKBLUE	0.854166667
14 All 07135019CHINOIS GREEN	0.851190476
50-DE All 065167191004 MILK	0.85
14 All 06906886BLACK	0.845238095
12 All 0708643442 DARK INDIGO - WORN IN	0.8375
14 All 0708643442 DARK INDIGO - WORN IN	0.833333333
5 All 0708643442 DARK INDIGO - WORN IN	0.829545455
26 All 07178209300 MARINE	0.794871795
30 All 07007601811 MANIC MIDNIGHT	0.794642857
14 All 071944538493 BROWN	0.794117647
12 All 07097745D223 SUPER BRIGHT STONED	0.793706294
10 All 06901351CLOUD DANCER/BLUE	0.791666667
12 All 06271997DARK ARMY	0.698830409
12 All 06272001DARK NAVY	0.697916667
30 All 07232538106 SCANDINAVIAN WHITE	0.697802198
34 All 07015867469 NAVY	0.592307692
37 All 07127470090 PURE WHITE	0.591666667
34 All 06031785017 SCHWARZ	0.590909091
26 All 07083284P23 MULTI/DARK CROSSHATCH DESTROYE	0.5875
50-DE All 072094321 BLACK	0.586538462
30 All 07135021CUBAN SAND	0.586111111
17 All 05543109811 MANIC MIDNIGHT	0.584795322

Accuracy Analysis on VisionWorks Data-Part 2(Planning item level)



Row Labels	Sum of Accuracy_ADIDA	Sum of Accuracy_imapa	Sum of Accuracy_tsb
AX	0.0%	-5.8%	57.4%
AY	-33.2%	-25.3%	-7.1%
BY	-45.7%	-32.2%	-62.4%
NEW LAUNCH	-18.4%	-7.1%	16.9%
Grand Total	-34.5%	-25.8%	-14.1%

MAPE | Retail Stores | Intermittency Flag : Yes



IntermittencyFlag	Yes										
Demand Class	Count of Intersection	Average of CoV	Sum of QTY	Periodicity	ZIR_sklego (lag 8)	Zir_sklego (lag 13)	imapa	tsb	ADIDA	o9_Stat	ZIR_LY
⊕ AY	21143	2.38	284876	1.26520295	1.234498316	1.286817	1.20123	1.2127	1.20097	1.37395	1.297617151
⊕ BY	11208	5.12	27029	1.41977718	1.373727661	1.394347	1.36394	1.383	1.36278	1.42341	1.399092274
⊕ NEW LAUNCH	1266	2.93	5183	1.46521853	1.374046884	1.454005	1.36412	1.3752	1.36417	1.1671	1.504831694
Grand Total	33617	3.32	317088	1.29790385	1.263670974	1.30976	1.23532	1.2483	1.23488	1.37268	1.319611696

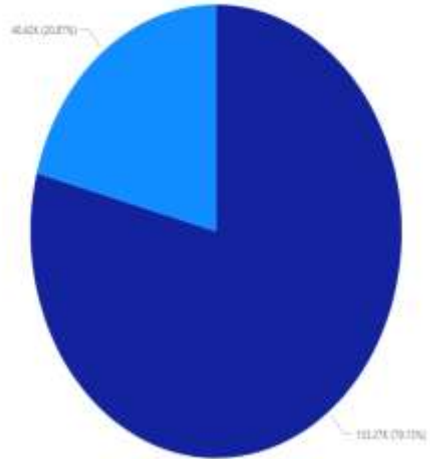
Observations :

- ZIR sklego and Periodicity models perform better than the DPRef Forecasting algorithms already present.
- ADIDA, iMAPA and TSB perform better than ZIR sklego and Periodicity Model.
- ZIR with bivariate LY flag feature has higher mape compared ZIR with lag 8 ang lag 13.

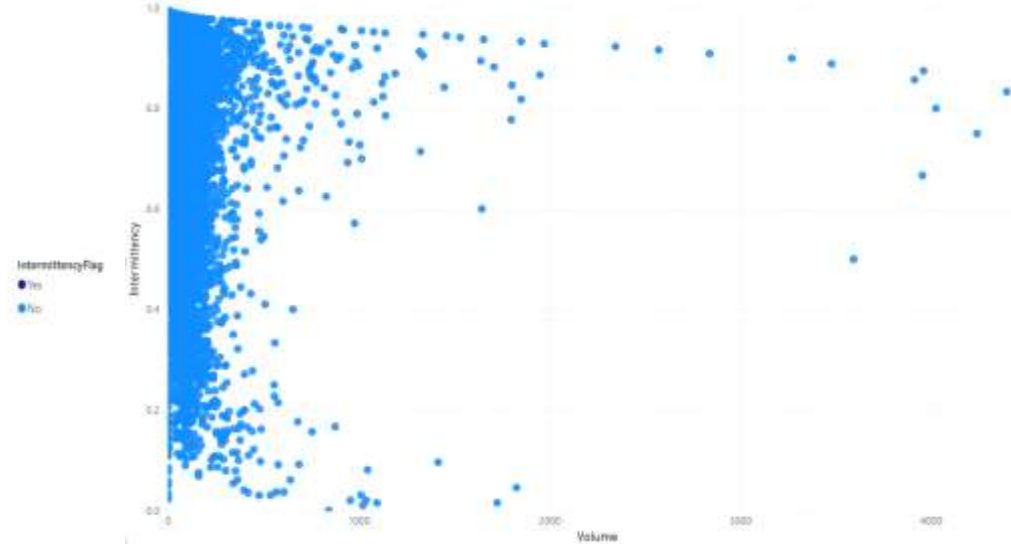
VisionWorks:Retail Stores



Intersection intermittency flag



Intermittency Distribution vs Volume



- So for VisionWorks the case study is conducted on Retail Stores as from the brief overview they have many intermittent time series.

Case Study - TWM



Column Name	Description
ZIR_Statsmodels_lag13	We have used lag 13 in regressor and library is statsmodels
ZIR_ExtraTreeRegressor_lag8	

tsid	ZIR_roc_auc
3-WIRE HARNESS FOR GO7_IOT_D95	85.83%
6 PIN HEAVY DUTY_IOT_D95	80.00%
ALCATEL 1 SE 32GB_DNE_41	50.00%
ALCATEL 1 SE 32GB_EMPRESARIAL_D39	75.00%
ALCATEL 1 SE 64GB_CALL CENTER_D83	75.00%
ALCATEL 1 SE 64GB_COR_EMPPL	50.00%
ALCATEL 1 SE 64GB_EMPRESARIAL_D39	71.43%
ALCATEL 1B COPPEL_COP	50.00%
ALCATEL 1B_COR_EMPPL	100.00%
ALCATEL 1B_EMPRESARIAL_D39	63.10%
ALCATEL 1B 5002M_CALL CENTER_D83	82.64%
ALCATEL 1B 5002M_COR_EMPPL	90.00%
ALCATEL 1B 5002M_DEPARTAMENTAL_LIV	50.00%
ALCATEL 1B 5002M_EMPRESARIAL_D39	62.50%
ALCATEL 1S 2019 COPPEL_COP	50.00%
ALCATEL 1V 2020 COPPEL_COP	50.00%
ALCATEL 3 2019_CALL CENTER_D83	81.25%
ALCATEL 3 2019 COPPEL_COP	50.00%



- AT&T - ADIDA, TSB, iMAPA, Ref Model Output, Croston Optimized, ZIR
 - Cycles - nov22, feb23
 - Forward forecast -
 - Significant lag - lag 1 (month)
 - Run time observations -
 - Accuracy/MAPE
 - Bias
 - Pick few good examples for reasoning
- VW -
 - WIP generations for few iterations
 - Models - ZIR, Ref Model Output, Periodicity, ADIDA, TSB, iMAPA, Croston Optimized

Case Study - VW





Compare with existing dpref performance.

- How Do we find significant lag.
- IY based features.



- We should have an approach to suggest the significant lag for any data set that is provided

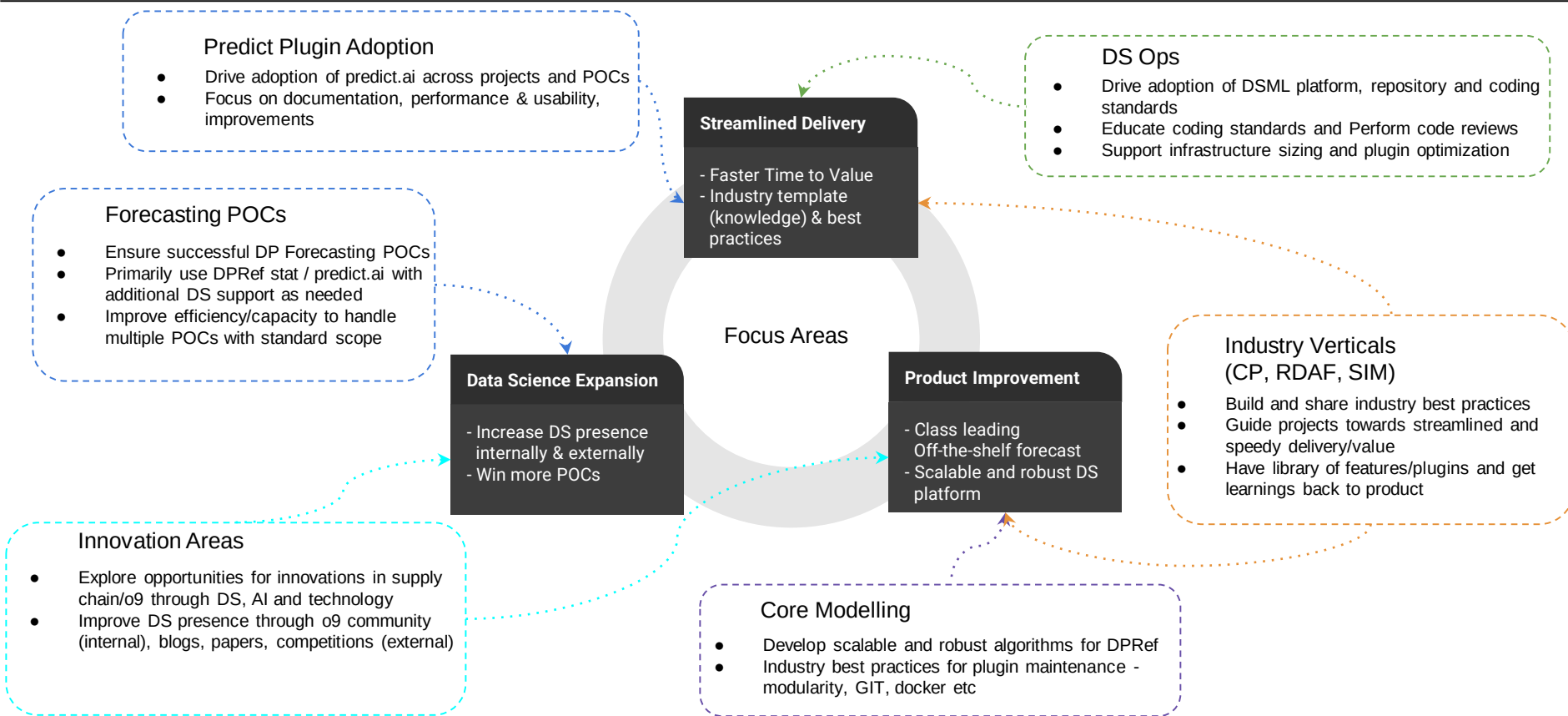


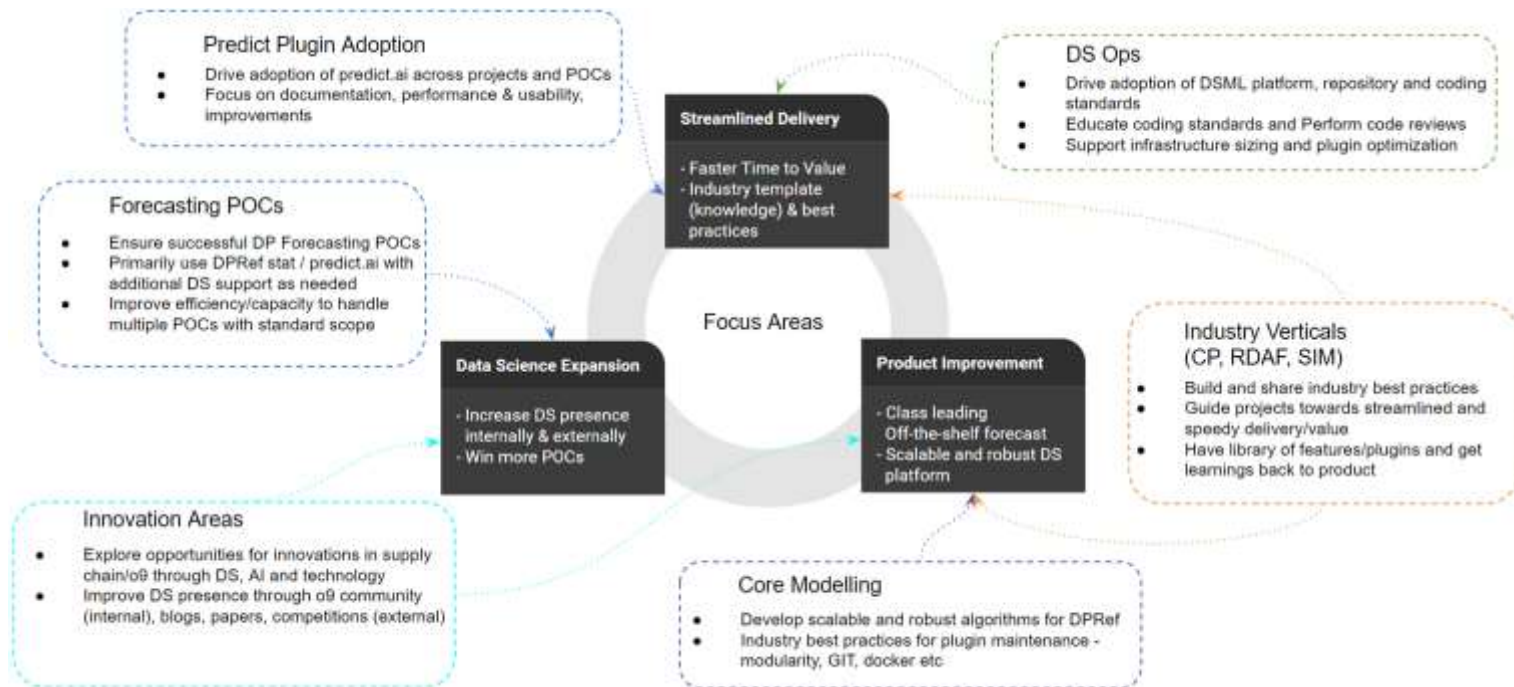
1. When forecast horizon is less than significant lags to be provided with some other default
 - a. Either set default lag equal to 3 or 4 or write some formula to





Data Science COE





09





	Short Term	Mid Term	Long term
1. Generate <i>Level</i> <i>Horizon</i> <i>Frequency</i>	Planning Item X Sold To X Day 1-42 Days Daily	Planning Item X Sold To X Wk 1-13 Weeks Weekly	Stat Item X Customer Group X Month 1-24 Months Monthly
2. Reconcile <i>Level</i> <i>Horizon</i> <i>Frequency</i>	TBD	Planning Item X Sold To X Week 1-13 week for visibility Every Monday	Stat Item X Customer Group X Week 1-24 Month as primary 1st day of the month
3. Override <i>Level</i> <i>Horizon</i> <i>Frequency</i>	No override	Limited	Review and Override
4. Publish <i>Level</i> <i>Horizon</i> <i>Frequency</i>	Weekly	Twice a month	Monthly

Short & Long-Term forecast

Use this format and add demand sensing in blue

	Short term: Deployment forecast	Long term: Demand forecast	Long term: Demand forecast
Decision	Drive short-term replenishment decisions	Drive long term capacity, sourcing needs and financial outlook/ gap closure	Drive long term capacity, sourcing needs and financial outlook/ gap closure
Frequency	Weekly	Monthly	Daily
Forecast Generation	Sell Out- Planning Item X Sold To X Week	Orders - Stat Item X Channel Group X Location Type X Month Sell Out - Stat Item X Customer Group X Month	Orders - Stat Item X Channel Group X Location Type X Month Sell Out - Stat Item X Customer Group X Month
Forecast Storage	Sell Out- Planning Item X Sold To X Week	Orders - Planning Item X Customer Group X Location X Week Sell Out - Planning Item X Customer Group X Week	Orders - Planning Item X Customer Group X Location X Week Sell Out - Planning Item X Customer Group X Week
Horizon	13 weeks	24 months	6-8 weeks
Algorithm	ML	Stat	Final Forecast Lag1
Target	Sellout	Orders and Sell Out	Final Deployment Forecast
Drivers	Promotions, Holidays	None	Open Orders, Shipments, Actualized
Reconciliation*	 Short term: Month 1-3  Long term: Month 4-24		

*Reconciliation Options are discussed in the next slide